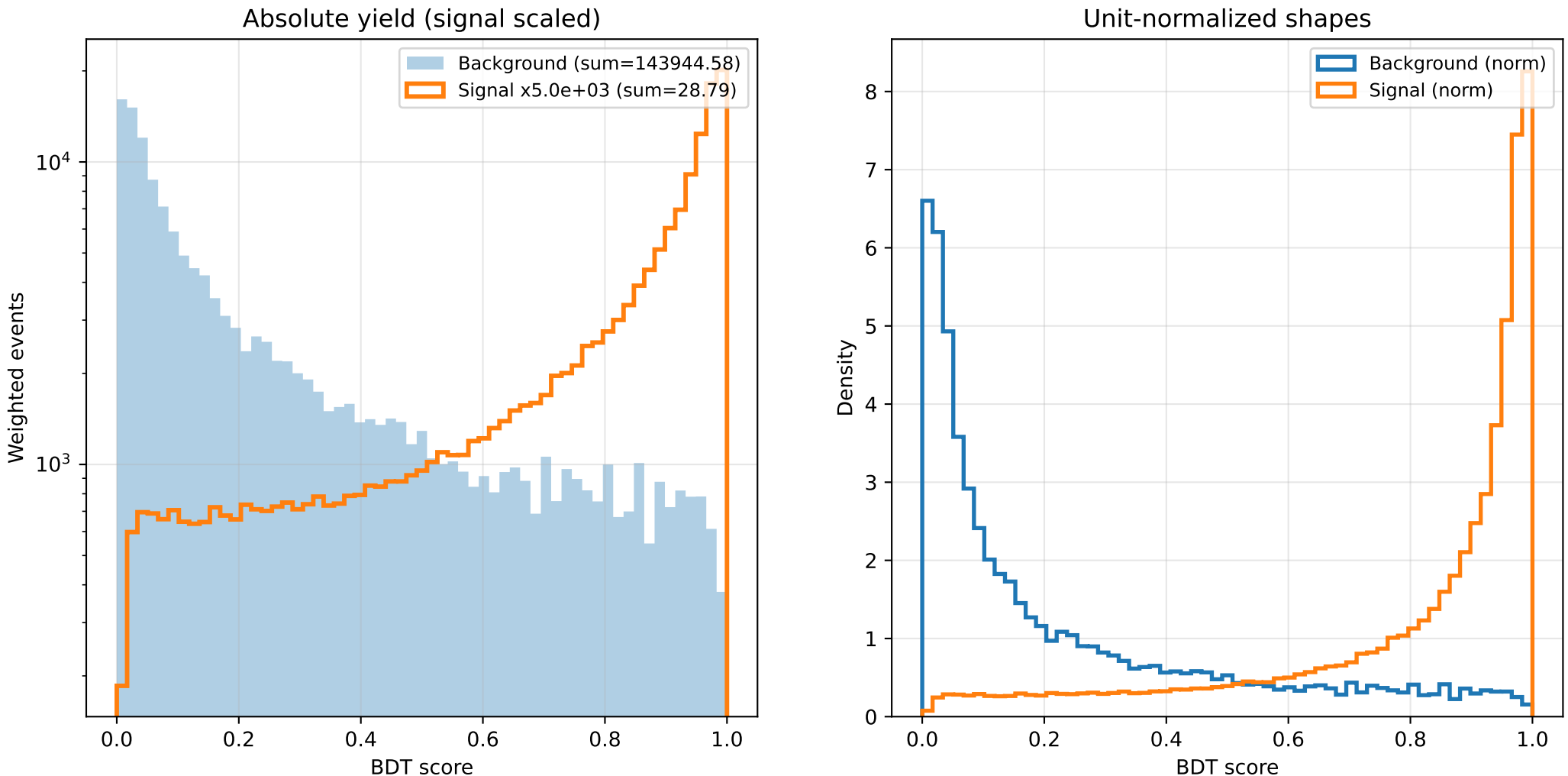
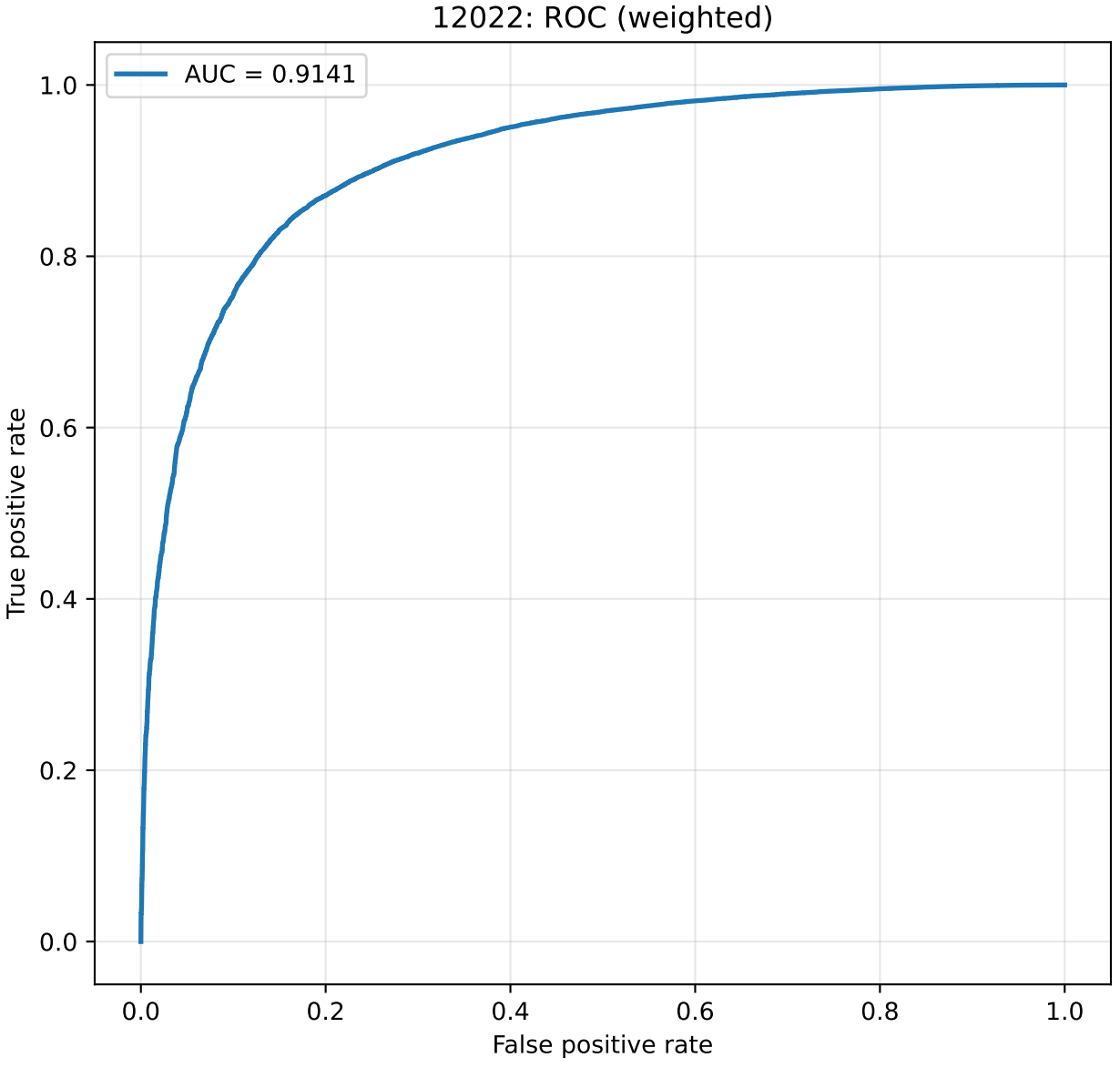
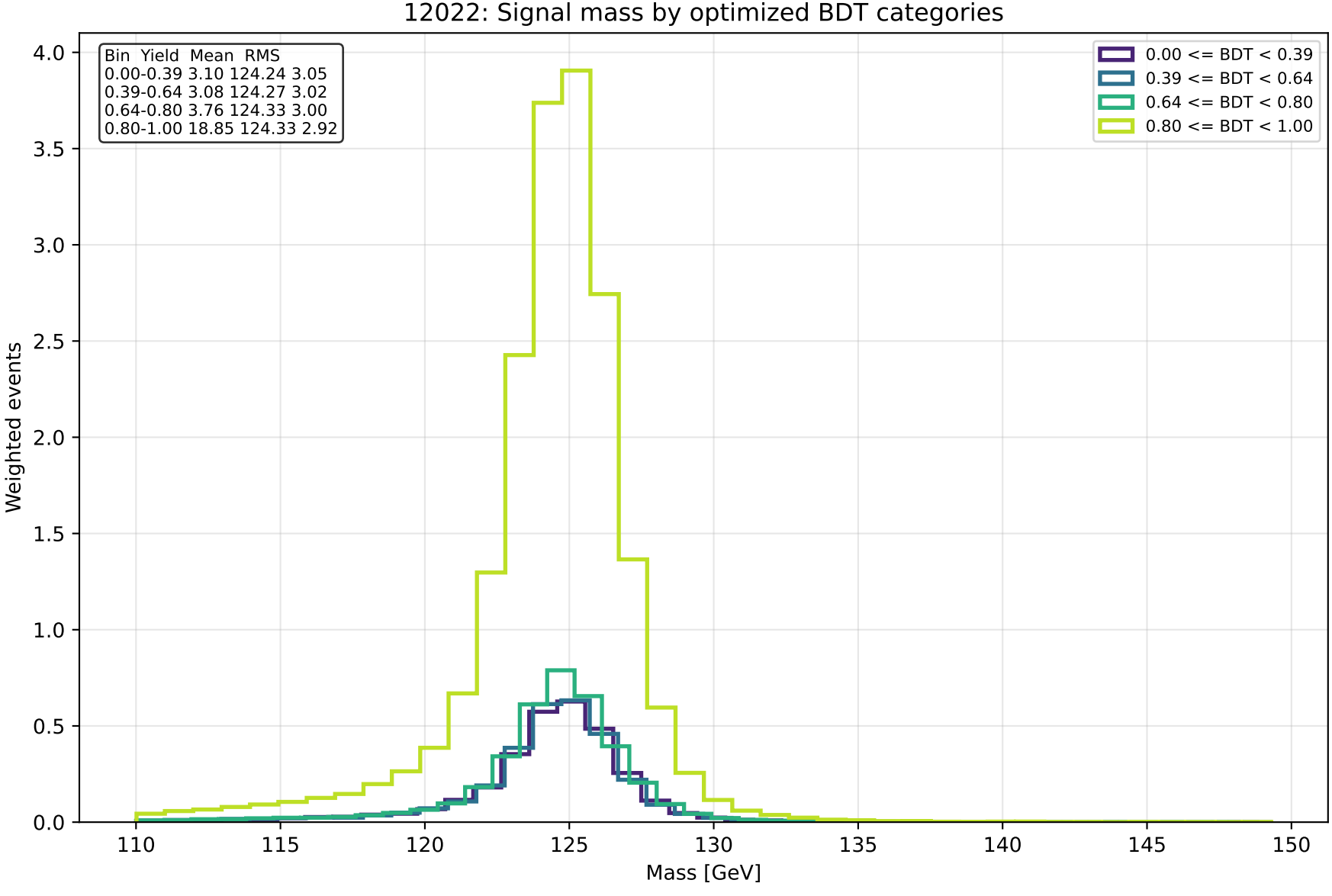
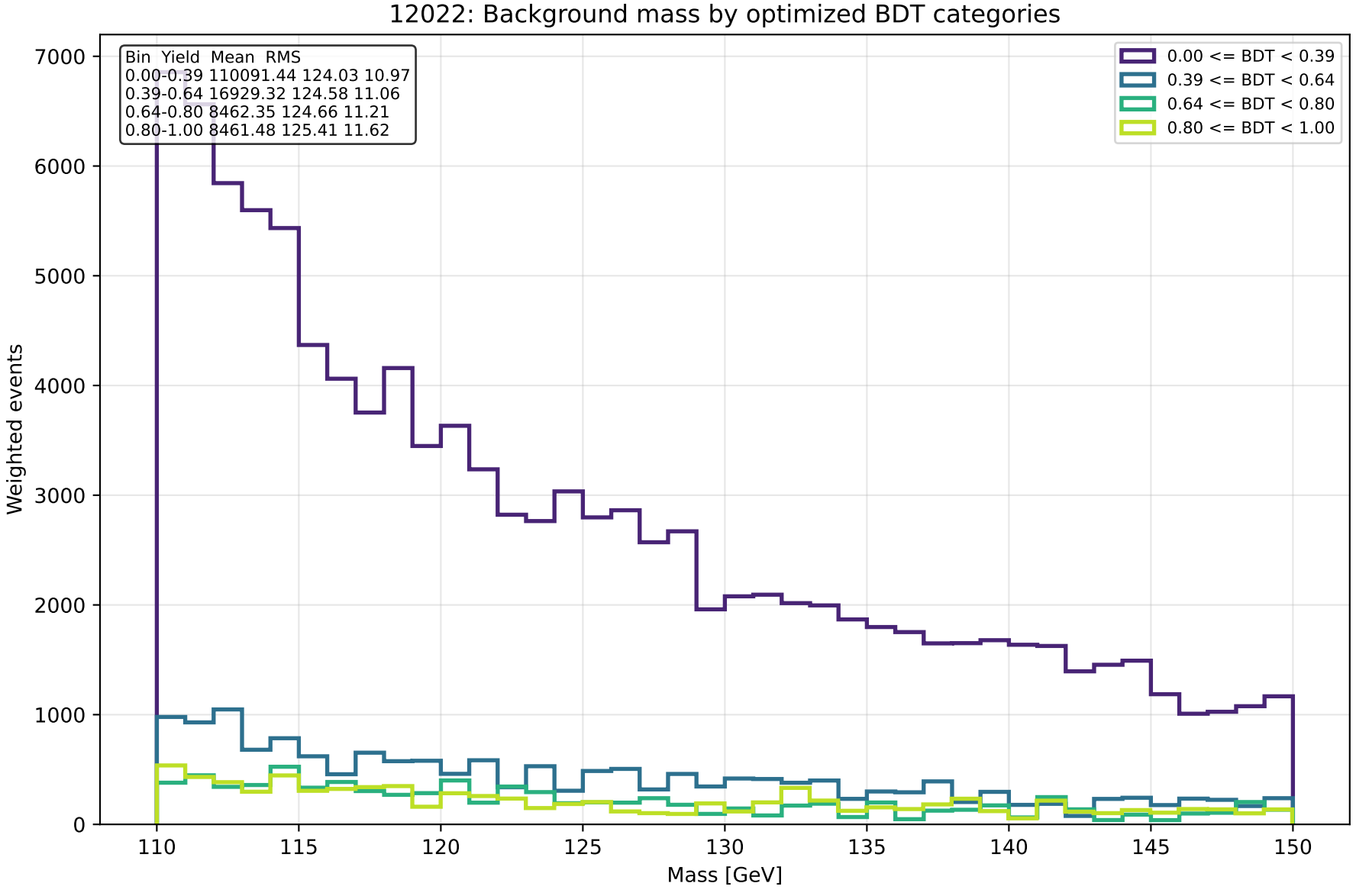

12022: BDT scores in 110-150 GeV
Absolute yield (signal scaled)
Background (sum=143944.58)
Signal x5.0e+03 (sum=28.79)
Weighted events
10^4
10^3
0.0
0.2
0.4
0.6
0.8
1.0
BDT score
Unit-normalized shapes
Background (norm)
Signal (norm)
Density
0
1
2
3
4
5
6
7
8
BDT score

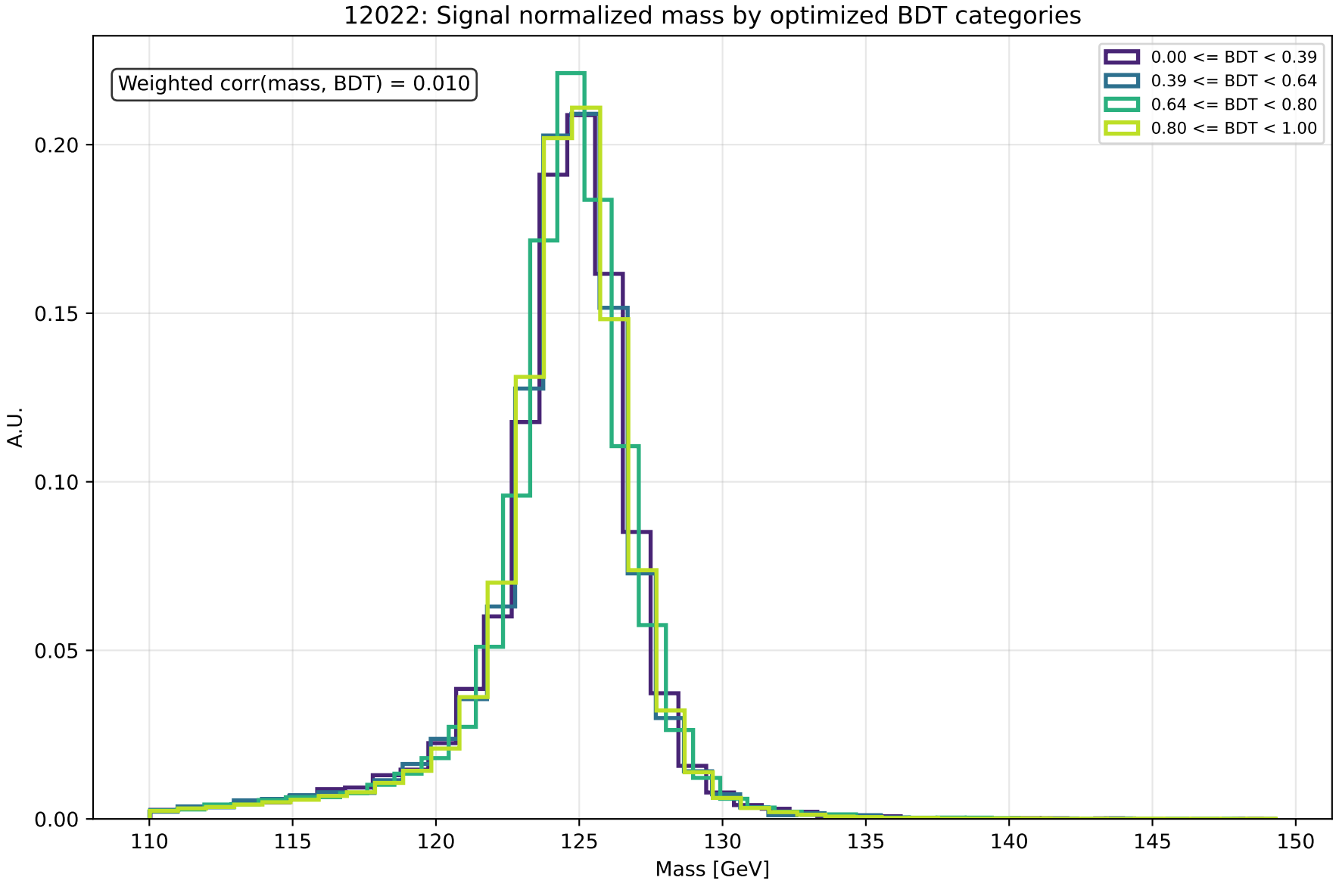

12022: ROC (weighted)
AUC = 0.9141
True positive rate
False positive rate
0.0
0.2
0.4
0.6
0.8
1.0

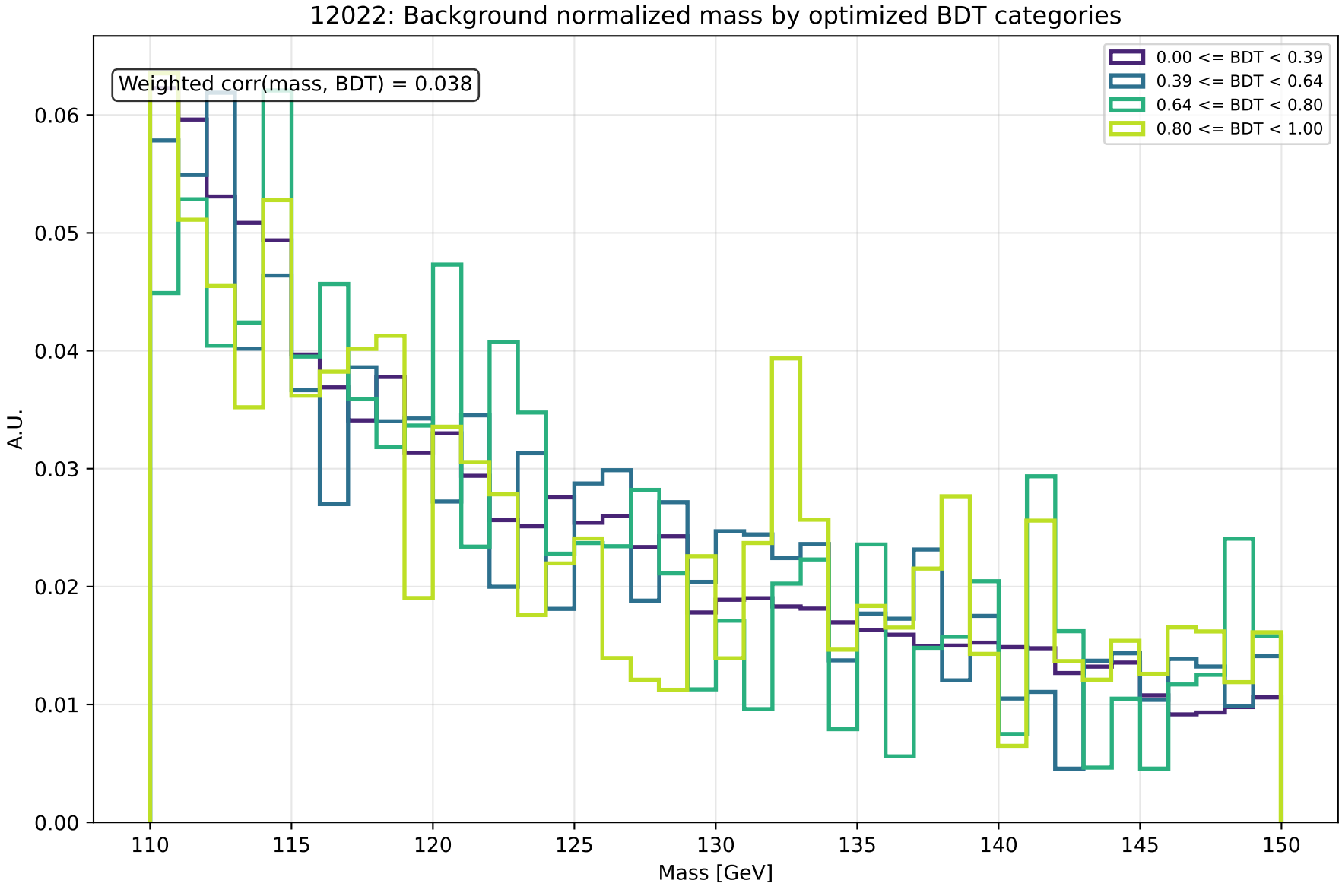
12022: Multi-bin BDT categorization (no hard cut)
Optimized N=4, Zexp(profile)=0.000, monotonic S/B=True

Bin	BDT range	S	B	S/B	S/sqrt(B)
1	0.000-0.386	3.097	110091.437	2.81e-05	0.009
2	0.386-0.641	3.083	16929.322	0.000182	0.024
3	0.641-0.804	3.765	8462.348	0.000445	0.041
4	0.804-1.000	18.848	8461.477	0.00223	0.205


12022: Signal mass by optimized BDT categories
Bin Yield Mean RMS
0.00-0.39 3.10 124.24 3.05
0.39-0.64 3.08 124.27 3.02
0.64-0.80 3.76 124.33 3.00
0.80-1.00 18.85 124.33 2.92
0.00 <= BDT < 0.39
0.39 <= BDT < 0.64
0.64 <= BDT < 0.80
0.80 <= BDT < 1.00
Weighted events
Mass [GeV]
0.0
0.5
1.0
1.5
2.0
2.5
3.0
3.5
4.0
110
115
120
125
130
135
140
145
150


12022: Background mass by optimized BDT categories
Bin Yield Mean RMS
0.00-0.39 110091.44 124.03 10.97
0.39-0.64 16929.32 124.58 11.06
0.64-0.80 8462.35 124.66 11.21
0.80-1.00 8461.48 125.41 11.62
0.00 <= BDT < 0.39
0.39 <= BDT < 0.64
0.64 <= BDT < 0.80
0.80 <= BDT < 1.00
Weighted events
Mass [GeV]
0
1000
2000
3000
4000
5000
6000
7000
110
115
120
125
130
135
140
145
150


12022: Signal normalized mass by optimized BDT categories
Weighted corr(mass, BDT) = 0.010
0.00 <= BDT < 0.39
0.39 <= BDT < 0.64
0.64 <= BDT < 0.80
0.80 <= BDT < 1.00
A.U.
0.00
0.05
0.10
0.15
0.20
110
115
120
125
130
135
140
145
150
Mass [GeV]

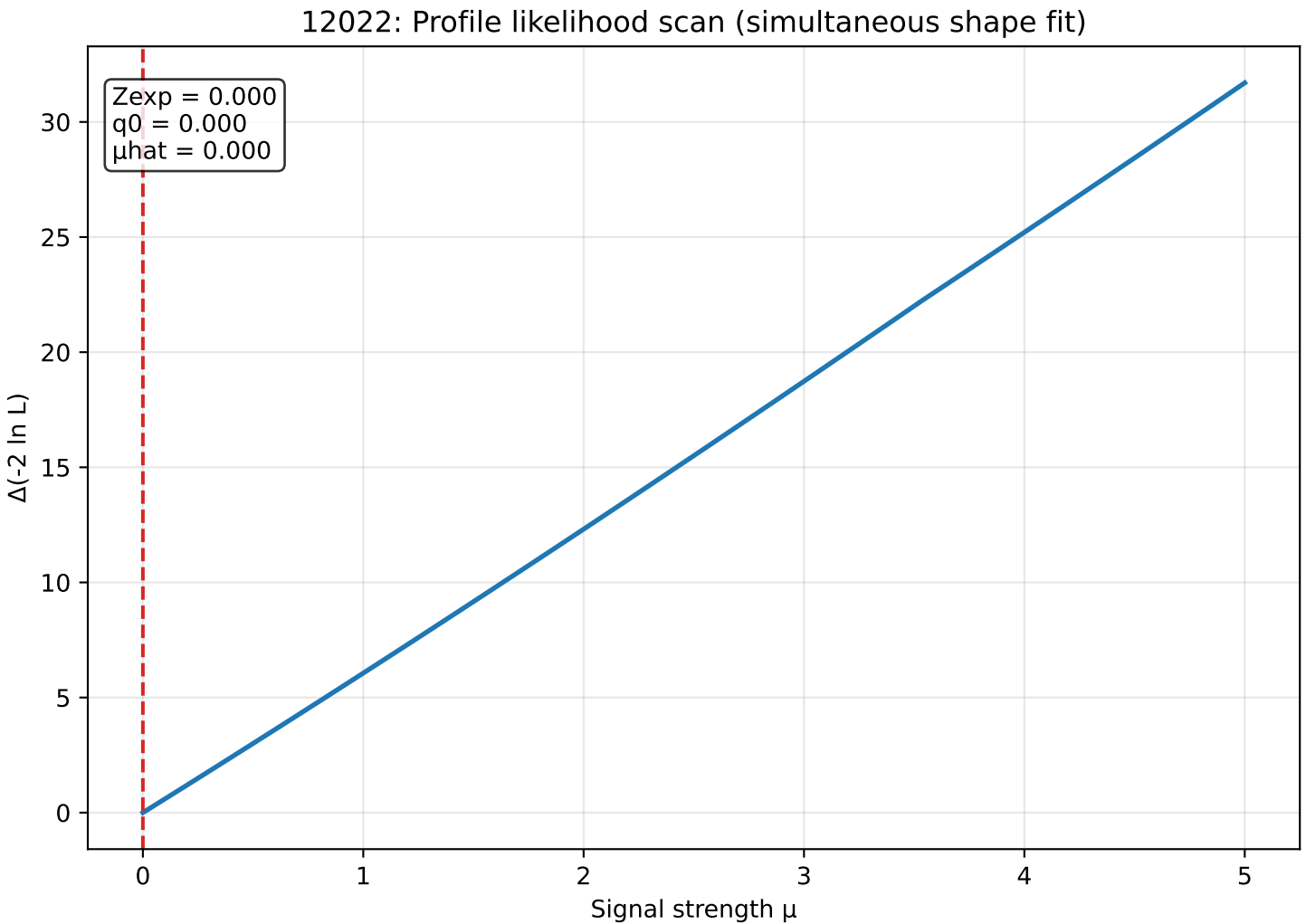

12022: Background normalized mass by optimized BDT categories
Weighted corr(mass, BDT) = 0.038
0.00 <= BDT < 0.39
0.39 <= BDT < 0.64
0.64 <= BDT < 0.80
0.80 <= BDT < 1.00
A.U.
0.00
0.01
0.02
0.03
0.04
0.05
0.06
110
115
120
125
130
135
140
145
150
Mass [GeV]

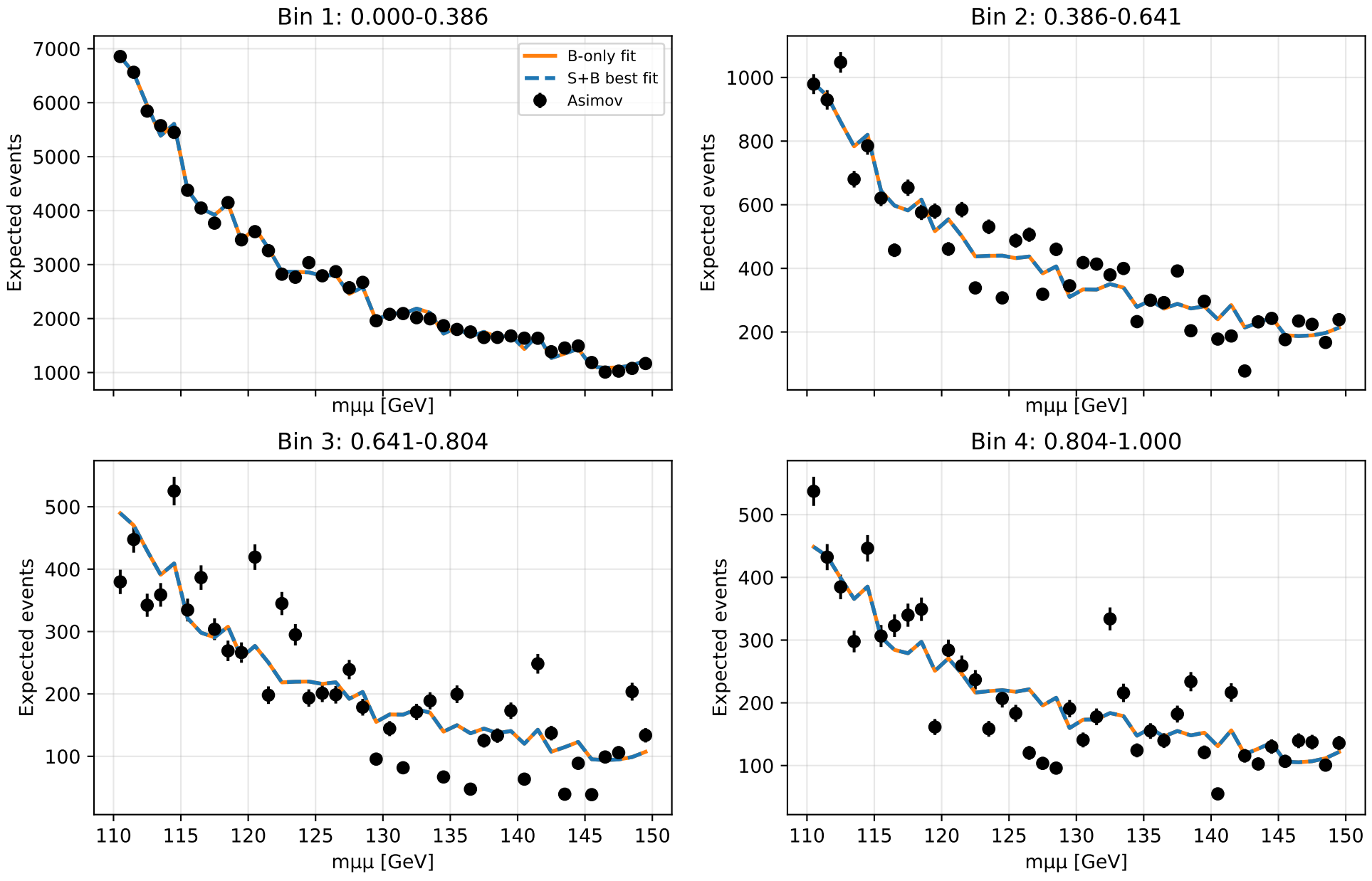
12022: Simultaneous shape-based profile likelihood summary

Quantity	Value
Expected Z	0.0000
q0	0.0000
mu_hat	0.0000
N_BDT bins	4
N_mass bins	40
poly order	1
bkg norm unc	0.200
poly unc	0.300
sig norm unc	0.050
sig mass shift unc [GeV]	0.200
sig resolution unc	0.100

12022: Shape-likelihood bin yields (Asimov)

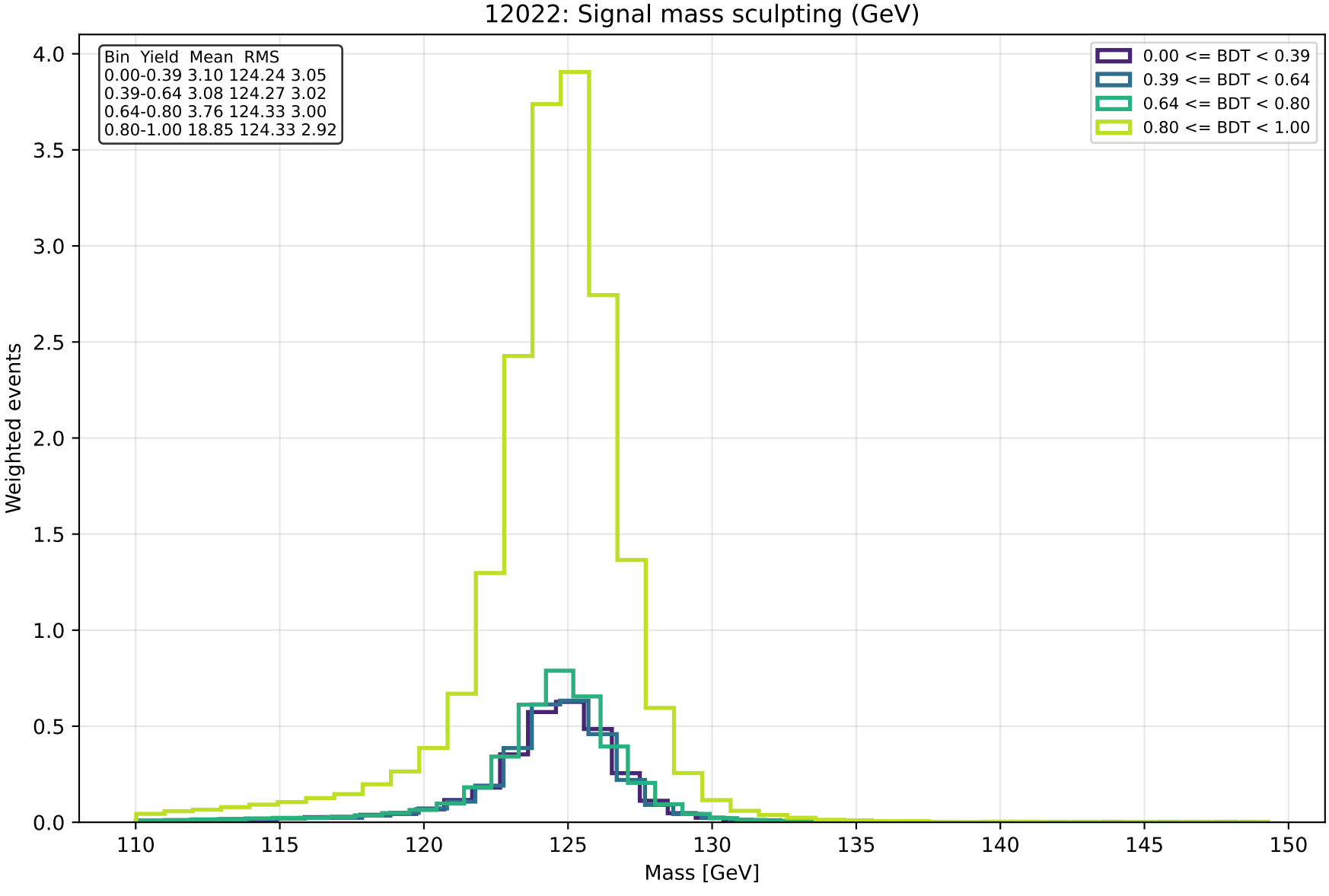
Bin	BDT	S truth	B truth	B-only fit	S+B fit
1	0.000-0.386	3.097	110091.437	110091.437	110091.437
2	0.386-0.641	3.083	16929.322	16929.322	16929.322
3	0.641-0.804	3.765	8462.348	8462.348	8462.348
4	0.804-1.000	18.848	8461.477	8461.477	8461.477

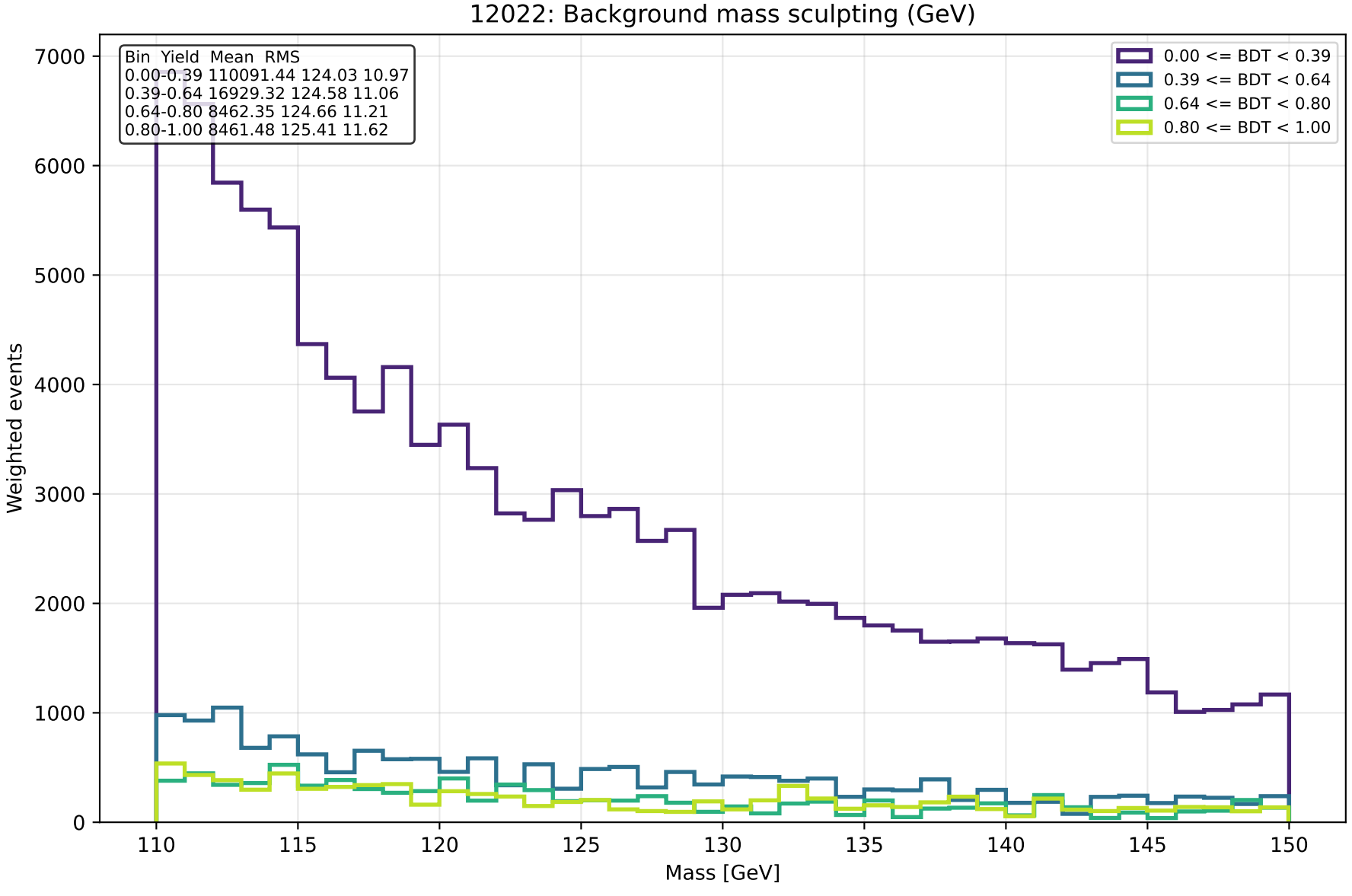

12022: Profile likelihood scan (simultaneous shape fit)
Zexp = 0.000
q0 = 0.000
μhat = 0.000
Δ(-2 ln L)
Signal strength μ
0
5
10
15
20
25
30
0
1
2
3
4
5

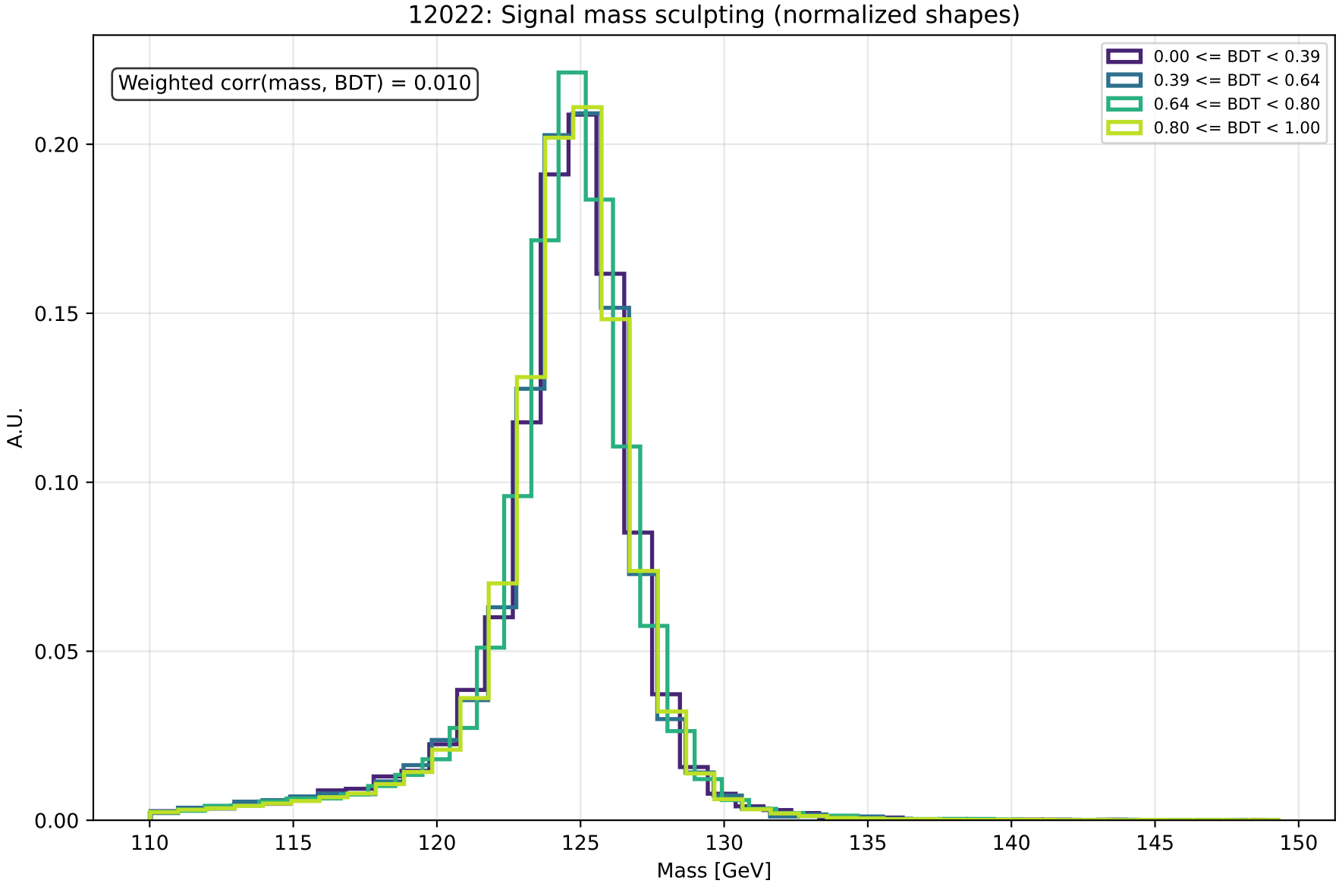

12022: Asimov mass templates by BDT bin (simultaneous fit)
Bin 1: 0.000-0.386
Bin 2: 0.386-0.641
Bin 3: 0.641-0.804
Bin 4: 0.804-1.000
B-only fit
S+B best fit
Asimov
Expected events
mμμ [GeV]

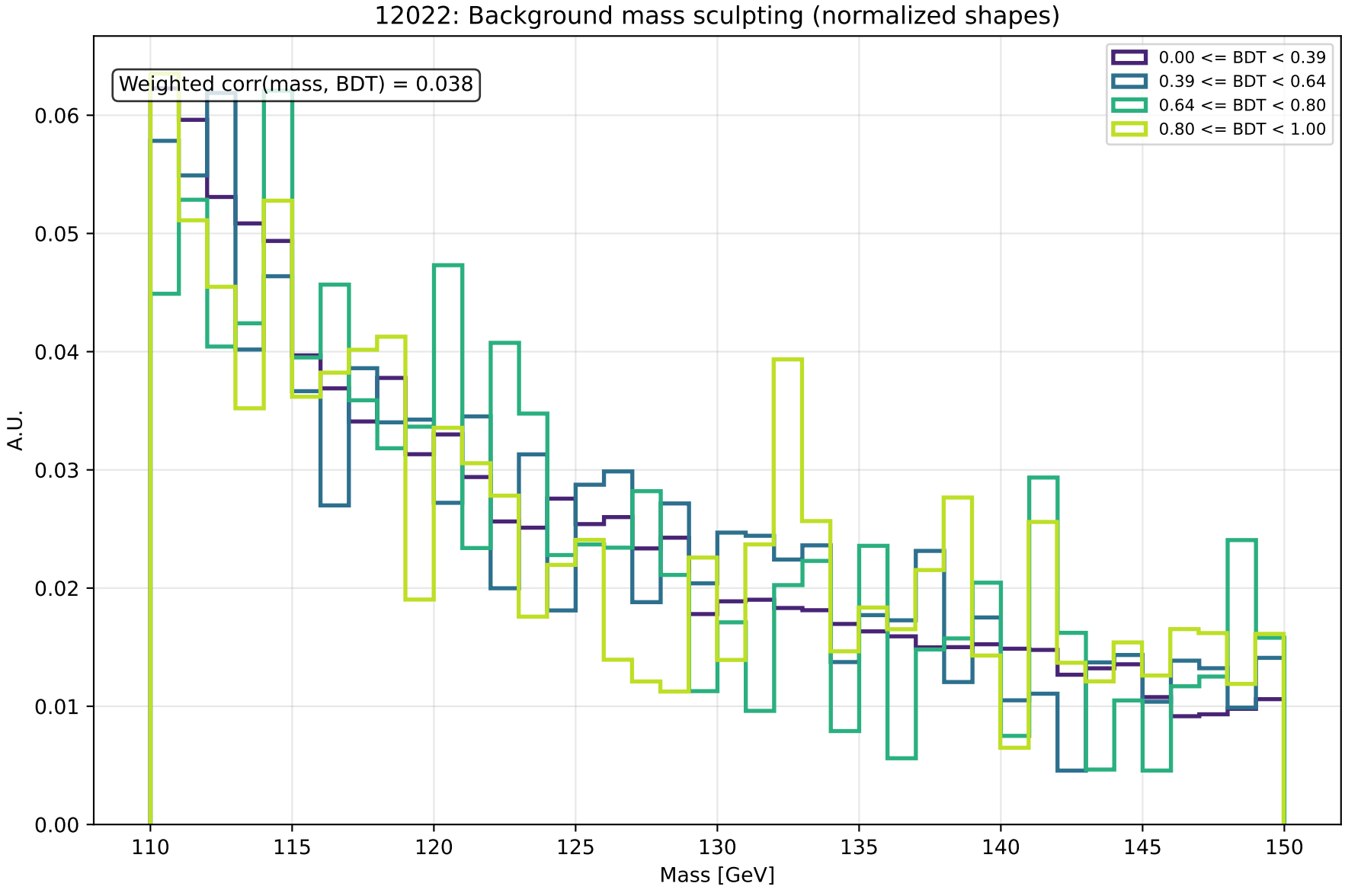
12022: Expected yield per optimized BDT bin (110-150 GeV)
S total=28.79, B total=143944.58

BDT bin	S (bin)	S frac	S>=lo	B (bin)	S/B (>=lo)
0.000-0.386	3.097	10.8%	100.0%	110091.437	0.0002
0.386-0.641	3.083	10.7%	89.2%	16929.322	0.000759
0.641-0.804	3.765	13.1%	78.5%	8462.348	0.00134
0.804-1.000	18.848	65.5%	65.5%	8461.477	0.00223

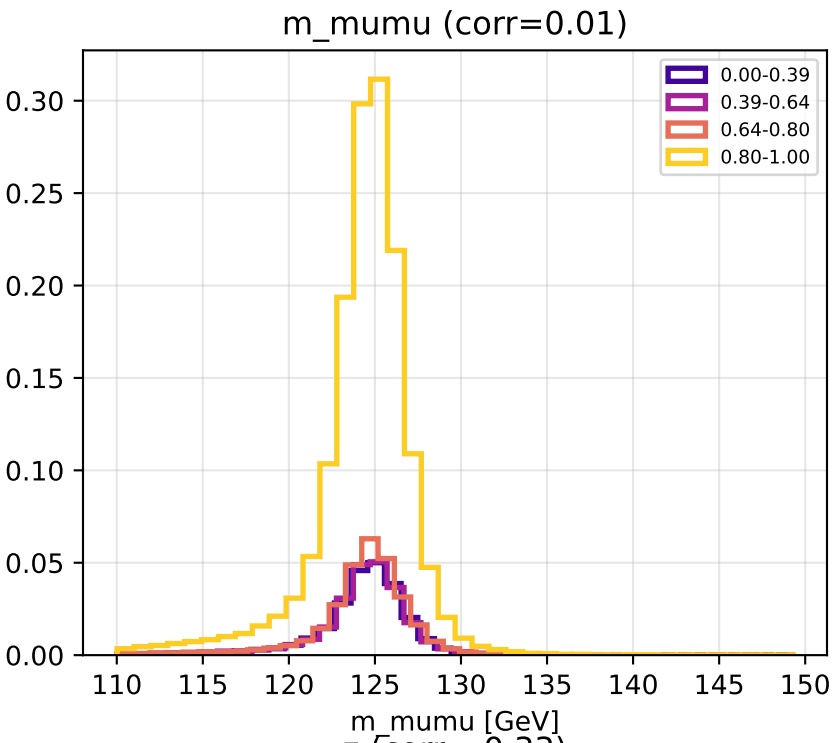

12022: Signal mass sculpting (GeV)
Bin Yield Mean RMS
0.00-0.39 3.10 124.24 3.05
0.39-0.64 3.08 124.27 3.02
0.64-0.80 3.76 124.33 3.00
0.80-1.00 18.85 124.33 2.92
0.00 <= BDT < 0.39
0.39 <= BDT < 0.64
0.64 <= BDT < 0.80
0.80 <= BDT < 1.00
Weighted events
Mass [GeV]
0.0
0.5
1.0
1.5
2.0
2.5
3.0
3.5
4.0
110
115
120
125
130
135
140
145
150

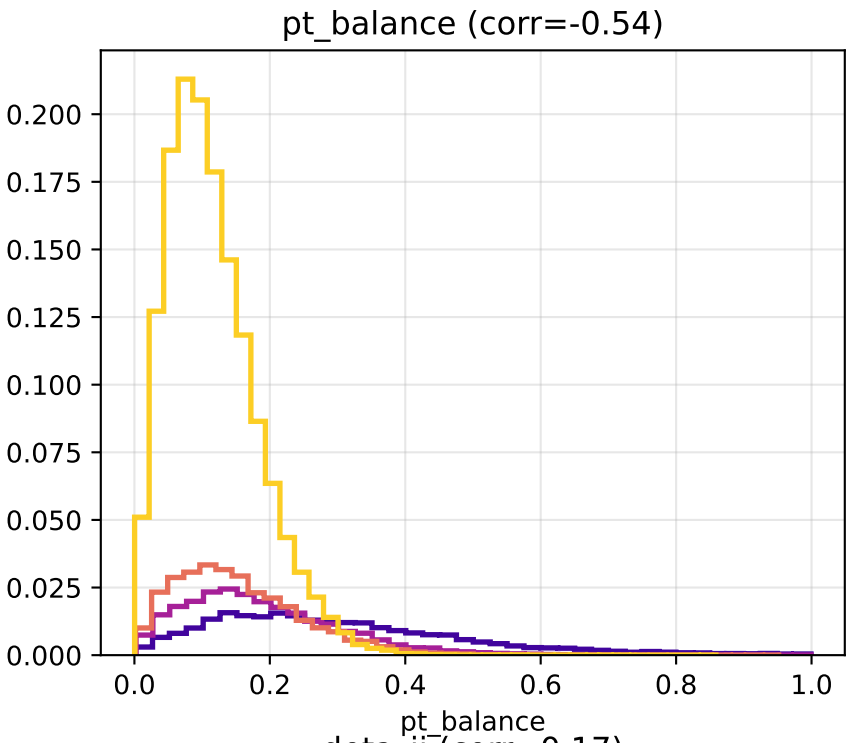

12022: Background mass sculpting (GeV)
Bin Yield Mean RMS
0.00-0.39 110091.44 124.03 10.97
0.39-0.64 16929.32 124.58 11.06
0.64-0.80 8462.35 124.66 11.21
0.80-1.00 8461.48 125.41 11.62
0.00 <= BDT < 0.39
0.39 <= BDT < 0.64
0.64 <= BDT < 0.80
0.80 <= BDT < 1.00
Weighted events
Mass [GeV]
0
1000
2000
3000
4000
5000
6000
7000
110
115
120
125
130
135
140
145
150

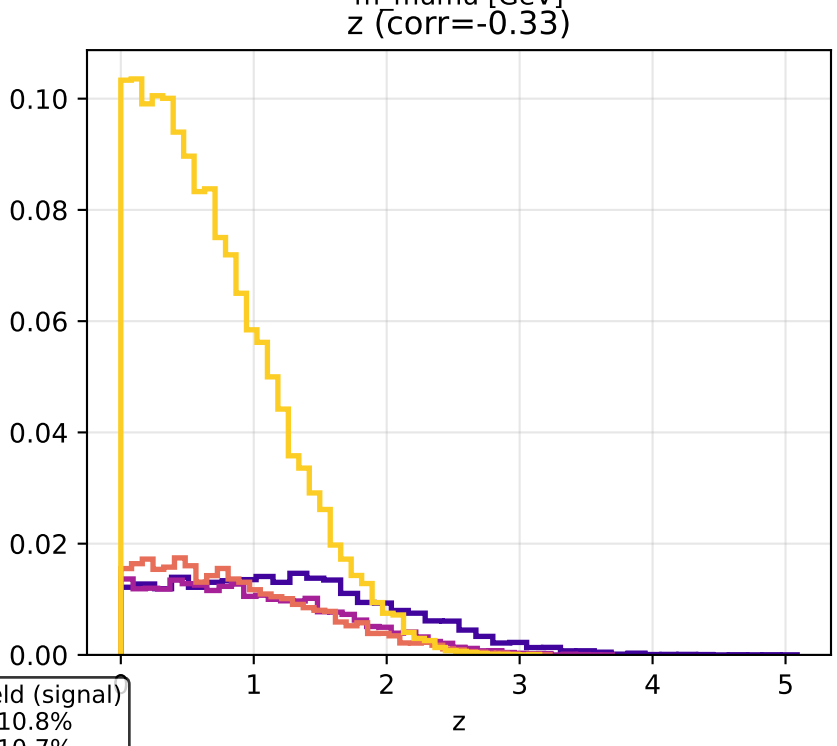

12022: Signal mass sculpting (normalized shapes)
Weighted corr(mass, BDT) = 0.010
0.00 <= BDT < 0.39
0.39 <= BDT < 0.64
0.64 <= BDT < 0.80
0.80 <= BDT < 1.00
A.U.
0.00
0.05
0.10
0.15
0.20
110
115
120
125
130
135
140
145
150
Mass [GeV]

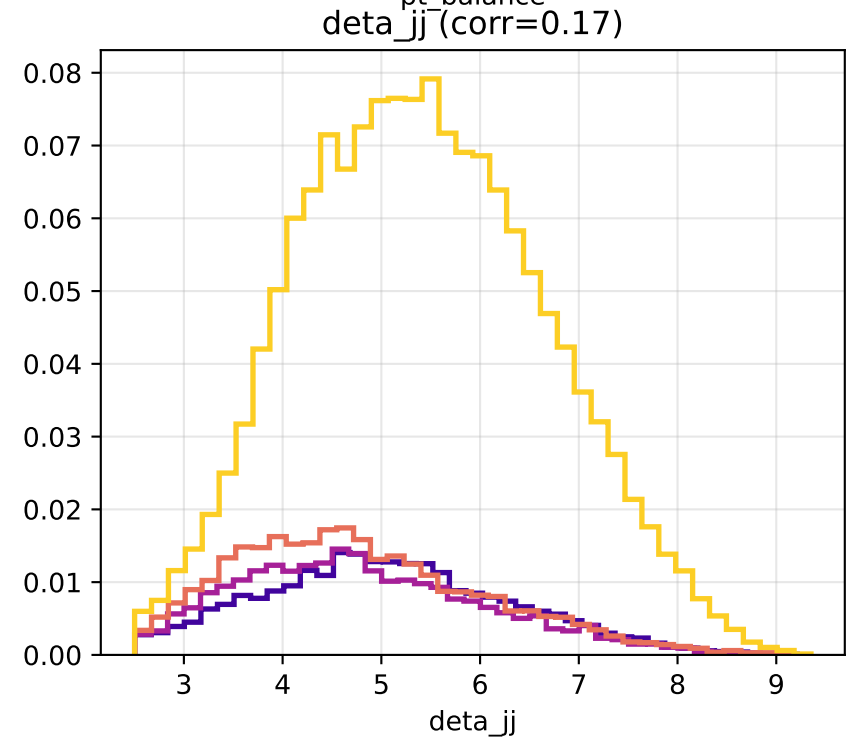

12022: Background mass sculpting (normalized shapes)
Weighted corr(mass, BDT) = 0.038
0.00 <= BDT < 0.39
0.39 <= BDT < 0.64
0.64 <= BDT < 0.80
0.80 <= BDT < 1.00
A.U.
0.00
0.01
0.02
0.03
0.04
0.05
0.06
110
115
120
125
130
135
140
145
150
Mass [GeV]

12022: Signal distributions by BDT bin

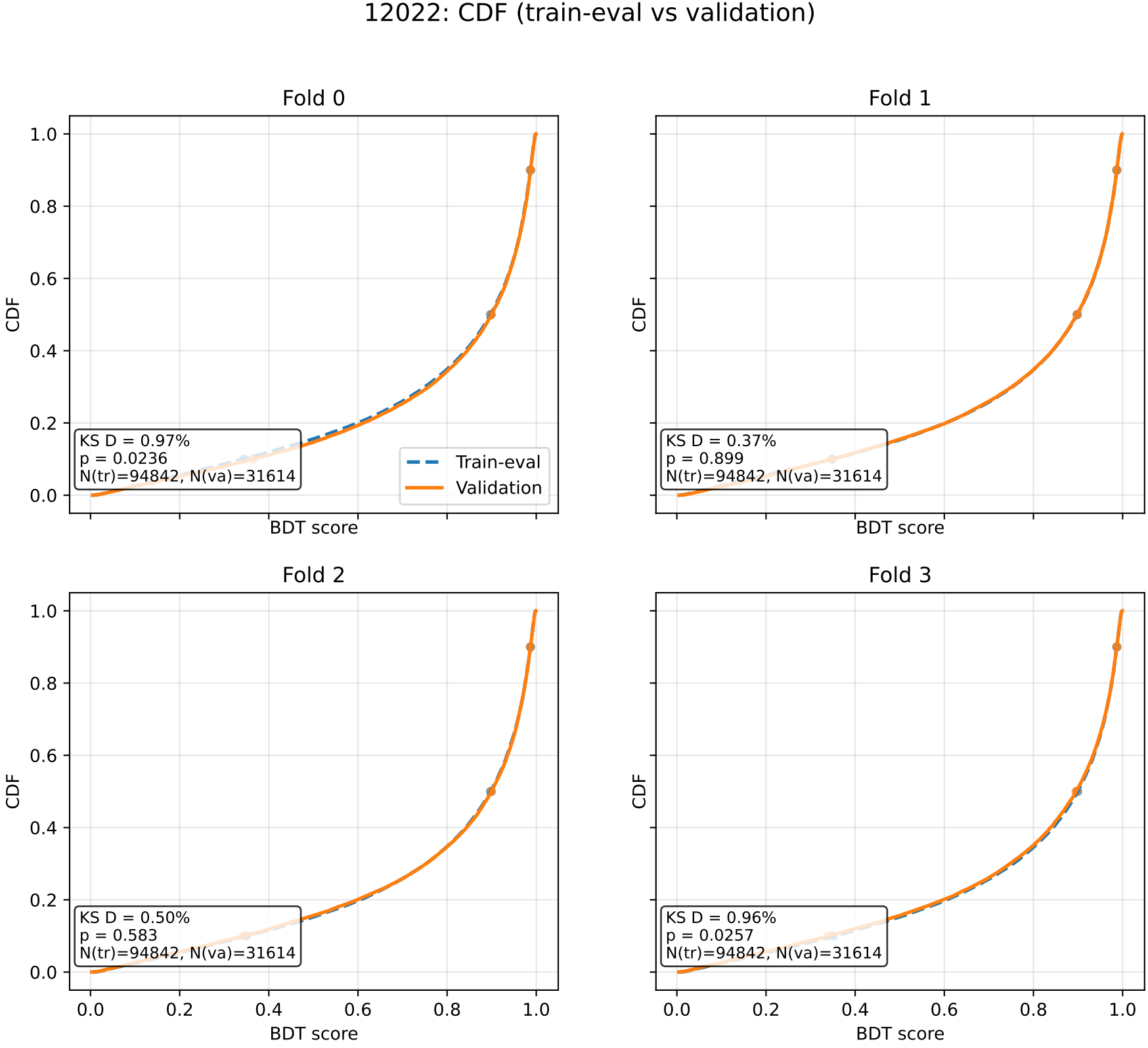








BDT bin yield (signal)
0.00-0.39: 10.8%
0.39-0.64: 10.7%
0.64-0.80: 13.1%
0.80-1.00: 65.5%


12022: CDF (train-eval vs validation)
Fold 0
KS D = 0.97%
p = 0.0236
N(tr)=94842, N(va)=31614
Train-eval
Validation
Fold 1
KS D = 0.37%
p = 0.899
N(tr)=94842, N(va)=31614
Fold 2
KS D = 0.50%
p = 0.583
N(tr)=94842, N(va)=31614
Fold 3
KS D = 0.96%
p = 0.0257
N(tr)=94842, N(va)=31614
CDF
BDT score
0.0
0.2
0.4
0.6
0.8
1.0

12022: KS test summary

Fold	KS statistic	N events	alpha	Pass threshold	KS pass
0	0.97%	31614	0.001	0.20%	FALSE
1	0.37%	31614	0.001	0.20%	FALSE
2	0.50%	31614	0.001	0.20%	FALSE
3	0.96%	31614	0.001	0.20%	FALSE

12022: Top-10 feature importance (gain)
12022_featureimportance_DY.txt

Feature	Gain %
pt_balance	11.97%
z	11.41%
deta_jj	10.34%
njets	9.72%
pt_j2	6.57%
dphi_mumu_jj	4.80%
jet1_eta	4.77%
costheta_CS	4.17%
pt_j1	4.06%
mu1_eta	4.01%